DSAI4205 Big Data Analytics

Group Project Report

[Project Title]

Text Classification Decision Support System

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Group number: DataVinci

Dataset used: 3
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Abstract

[Abstract text]

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1 Introduction and Dataset

1 1 Background and decision problem

[Background, intended users, decision context and motivation]

1 2 Project objectives

[Classification task, target labels, scope and measurable success criteria]

1 3 Dataset description

[Dataset provenance, collection context, label origin and reference]

Table 1 Dataset overview

Attribute	Description
Dataset and version	[Dataset name and version]
Source and time coverage	[Source citation and period]
Unit of analysis	[Meaning of one record]
Size and language	[Number of records and language]
Target labels	[Class names and definitions]
Usage conditions	[Relevant access or usage conditions]

Table 2 Data dictionary

Field	Type	Description
[Text field]	[Type]	[Meaning and role]
[Target field]	[Type]	[Meaning and role]
[Identifier or group]	[Type]	[Meaning and role]

2 Exploratory Data Analysis

2 1 Data quality and descriptive statistics

[Summary of missing values, duplicates, label quality and text characteristics]

Table 3 Data quality summary

Measure	Value	Interpretation
Raw records	[N]	[Observation]
Missing text or labels	[N / %]	[Observation]
Duplicates	[N / %]	[Observation]
Median text length	[Value and unit]	[Observation]

2 2 Class distribution

[Insert figure]

Figure 1 Class distribution

[Interpretation of class balance and implications for evaluation]

2 3 Text characteristics

[Insert figure]

Figure 2 Text length distribution

[Patterns in text length, vocabulary or representative examples and implications for preprocessing]

3 Preprocessing and Methodology

3 1 Data cleaning

Table 4 Data cleaning operations

Operation	Rationale	Records affected
[Operation]	[Reason]	[N]
[Operation]	[Reason]	[N]
Final retained dataset	[Inclusion criteria]	[N]

[Cleaning sequence, before and after examples, and treatment of conflicting labels]

3 2 Experimental design

Table 5 Data partitioning

Partition	Records	Proportion	Purpose
Training	[N]	[%]	Model and feature fitting
Validation	[N]	[%]	Model selection
Test	[N]	[%]	Final evaluation

[Split strategy, random seed, grouping or time constraints, cross validation if used, and leakage prevention]

3 3 Text representation

[Tokenization, normalization, representation method and fitted feature settings]

3 4 Baseline and candidate models

[Reference baseline, candidate algorithms, hyperparameters, tuning procedure and model selection rule]

3 5 Evaluation and computing environment

[Primary and secondary metrics with rationale, runtime measurement method, hardware and software versions]

4 Results and Error Analysis

4 1 Model comparison

Table 6 Validation performance and computational cost

Model	Accuracy	Macro F1	Training time
[Reference]	[Value]	[Value]	[Value / unit]
[Baseline]	[Value]	[Value]	[Value / unit]
[Candidate]	[Value]	[Value]	[Value / unit]

[Comparison on the same evaluation protocol and justification for the selected model]

4 2 Final test performance

Table 7 Selected model performance by class on the test set

Class	Precision	Recall	F1	Support
[Class]	[Value]	[Value]	[Value]	[N]
[Class]	[Value]	[Value]	[Value]	[N]
Macro average	[Value]	[Value]	[Value]	[N]

[Insert figure]

Figure 3 Confusion matrix for the selected model on the test set

[Overall test metrics, class level findings and uncertainty or stability assessment]

4 3 Error analysis

Table 8 Representative classification errors

Case	True label	Predicted label	Explanation
[ID]	[Label]	[Label]	[Excerpt and likely cause]
[ID]	[Label]	[Label]	[Excerpt and likely cause]

[Recurring error patterns, their impact and potential remedies]

5 Discussion and Conclusion

5 1 Interpretation of findings

[What the results establish, why the selected approach performs as observed, and how findings address the objectives]

5 2 Decision support workflow

[Insert figure]

Figure 4 Decision support workflow

[Intended user interaction, model output, resulting action, review process, and implemented or proposed system components]

5 3 Business implications

[Evidence based value, consequences of false positives and false negatives, and assumptions behind any projected benefit]

5 4 Scalability and practical feasibility

[Measured data scale, processing time and resource requirements; implications for larger workloads and operational use]

5 5 Limitations and future work

[Limitations of data, labels, evaluation and generalization; relevant privacy or bias considerations; prioritized improvements]

5 6 Conclusion

[Answer to the project objective, principal supported results and practical recommendation]

References

[Author or organization. (Year). Title. Publisher or source. URL or DOI.]

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Appendix A Reproducibility

Table A1 Reproducibility record

Item	Details
Submitted notebook	[Filename and entry point]
Dataset version	[Version and access reference]
Environment	[Runtime and package versions]
Execution	[Steps to reproduce results]
Randomness and splits	[Seeds and split identifiers]
Experiment configuration	[Saved parameters and model settings]

Appendix B Supplementary Results

[Additional metrics, parameter settings, robustness checks, figures or examples referenced in the main report]